

SANJEEVINI AUSHAD

AI-Powered Medication Allergen Safety — a Clinical Decision Support System

The remedy, brought in time.

33M

Americans with food allergies

44.86%

of 42,052 oral medications contain dairy

60%

of adult anaphylaxis historically 'idiopathic'

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My Story — The Patient Nobody Connected

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March 2025. Severe abdominal pain — so debilitating I could barely move.

- My dairy anaphylaxis was documented everywhere — MyChart, pharmacy records, my allergist's notes.
- A medication containing a dairy-derived excipient was still dispensed to me. No system checked.
- I suffered from March to July — physically, mentally, and financially.
- The insurance claims came to roughly \$30,000 — PCPs, GI, allergist, procedures.
- I found the cause myself, by reading ingredient lists nobody else had read.

The larger truth

The data was always there. The system simply failed to connect it. Patients are left to navigate dangerous gaps between EHRs, prescribers, and pharmacies alone.

4+

months of avoidable suffering

\$30K

in insurance claims

The Problem — Allergens Hiding in Medications

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Medications routinely contain top-9 food allergens as inactive ingredients. Nobody is checking.

- Lactose and starch are common binders; soy and egg derivatives appear across drug classes.
- Prescribers and pharmacists have no tool that cross-checks excipients against a patient's documented allergies.
- Patients are told to “call the manufacturer” — a toll-free dead end.
- Reactions get labeled “idiopathic” because nobody checks the ingredient list.
- Labeling law hasn't caught up: the ADINA Act (born from Adina Togal's reaction to an unlisted allergen in her antibiotic) is proposed — not yet law.

One allergen, many names — milk alone:

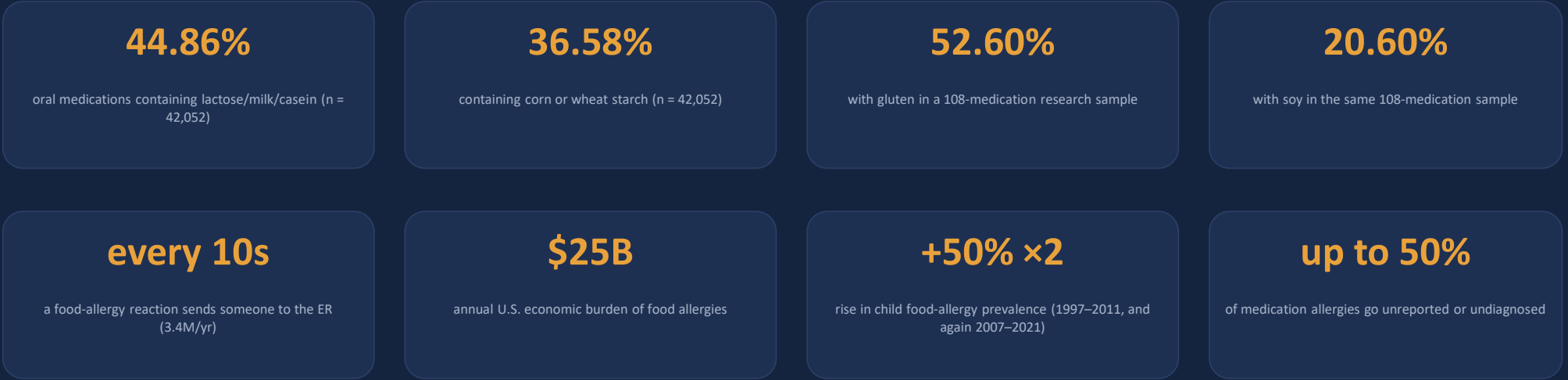
Casein	Caseinate	Whey	Lactose	Lactalbumin
Lactoglobulin	Milk solids	Ghee	Cream	
Nonfat milk solids	Whey powder	Lactulose		

Self-advocacy means memorizing all of these — for every allergen, on every prescription, forever.

How Bad Is It? — The Numbers

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From my initial data collection (openFDA drug labels) and published research.



The exposure is not rare — it is routine. And it is under-recognized by patients and providers alike, because no standardized measure of allergen prevalence in medications exists. That measurement gap is where this project begins.

Current Solutions — Manual, Complex, Error-Prone

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Today the entire burden of safety falls on the patient.

Self-advocacy

Patients must research every prescription themselves — across 12+ synonyms per allergen — with fragmented, unmaintained online resources.

Ask the manufacturer

A toll-free number that rarely produces a definitive, timely answer — while the prescription waits.

EHR & pharmacy systems

Do not routinely flag inactive ingredients against documented allergies at prescribing or dispensing. The data is not connected.

The ADINA Act

Would mandate allergen labeling in medications — an important step, but not yet law, and labels alone don't check YOUR allergy list.

Why are we doing all of this manually when the ingredient data is public, structured, and machine-readable?

Why This Is Solvable Now — The Data Already Exists

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Every ingredient of every U.S. medication is public and machine-readable. Nobody connects it to patients.

openFDA

Complete drug labels — active & inactive ingredients — updated monthly, free API

DailyMed

FDA Structured Product Labels (SPL) for every marketed drug

RxNorm

Standardized drug naming and identifiers to unify sources

EHR allergy lists

The patient's documented allergies — already in MyChart et al.

All four exist today. The missing piece is the connection: a system that reads the ingredient data, understands allergen synonyms, and checks it against the patient's own allergy list — automatically, at the moment it matters.

Formulations change monthly — which is why this must be a living system, not a one-time report.

The Solution — Three Parts, One System

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Named for what each part does: see the problem, draw the line, bring the remedy.



Divya Drishti

"Divine Sight"

SEE how bad the problem is: descriptive analysis of allergen prevalence across all FDA-approved oral medications — refreshed monthly.



Lakshmana Rekha

"The Protective Line"

STOP unsafe medicines at the point of prescribing: cross-check the patient's allergy list against every ingredient. Alert, explain, and hold the line.



Sanjeevini

"The Remedy"

OFFER a safe path forward: a therapeutically equivalent, allergen-free alternative — filtered to what the patient's insurance supports.

Descriptive → Prescriptive → Restorative. Each part stands alone; together they close the loop from evidence to protection to care.

Part 1 — Divya Drishti: How Bad Is the Problem?

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Problem type: Descriptive Indicators. You cannot fix what you have not measured.

- Compute the exact percentage of current FDA-approved oral medications containing one or more of the top-9 allergens.
- Mode: identify the most common culprit excipient (e.g., lactose as a binder).
- Frequency distributions: which manufacturers use allergen excipients most.
- Published as a monthly refreshed public report — because openFDA data and drug formulations change every month.

Why it matters

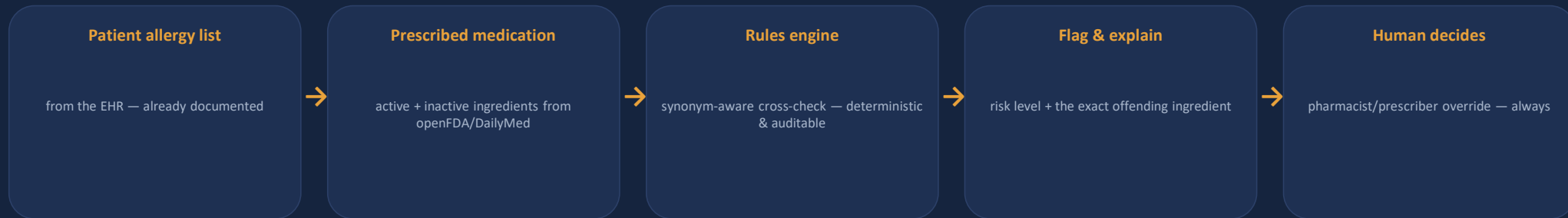
Without a baseline, this problem stays invisible — dismissed case by case as “idiopathic.” A single credible number (“X% of oral medications contain a top-9 allergen”) gives patients, providers, advocacy groups, and policymakers something to act on. Divya Drishti is the sight that makes the invisible undeniable.

Data: openFDA drug labels · allergen synonym dictionary · automated monthly pipeline

Part 2 — Lakshmana Rekha: The Line No Unsafe Medicine Crosses

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Problem type: Prescriptive. Stop the harm at the moment of prescribing — not after the reaction.



- Delivered where the decision happens: EHR plugin (Epic / Cerner), pharmacy system, or standalone web/mobile check.
- Deliberately rule-based at the core: for safety decisions, deterministic and auditable beats black-box — every flag can be explained.
- Human-in-the-loop by design: the system informs and warns; the clinician always holds final authority.

In the Ramayana, the Lakshmana Rekha was a protective line that kept danger out. This is that line, drawn in software, around every allergic patient.

Part 3 — Sanjeevini: Don't Just Say No — Offer the Remedy

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A stop without an alternative just moves the problem. Sanjeevini completes the care loop.

- When Lakshmana Rekha flags a medication, Sanjeevini searches for a therapeutically equivalent product without the patient's allergen.
- Same active ingredient, different excipients — the safe formulation often already exists.
- Filtered by the patient's insurance formulary, so the alternative is affordable and actionable — not just theoretically safe.
- Presented to the prescriber as a suggestion, never an automatic switch.

Example — hypothyroid patient

Synthroid

levothyroxine

Contains lactose monohydrate as an excipient — a problem for dairy-allergic patients.



e.g., Levoxyl

levothyroxine

Same molecule, lactose-free formulation — verified against current labeling, insurance-supported.

In the Ramayana, the Sanjeevini herb saved a life because it arrived in time. The safe alternative usually exists — the system's job is to bring it to the prescriber in time.

The hard part is not the rules engine. It is making messy, monthly-changing data trustworthy.

Heterogeneous sources

openFDA JSON, DailyMed XML/SPL, unstructured label text — one pipeline must normalize all of them.

Synonym resolution

12+ names for milk alone; every top-9 allergen has aliases. A curated, versioned synonym dictionary is the core asset.

Identifier chaos

The same drug appears under many names and NDCs — RxNorm standardization required before any cross-check.

Monthly drift

Formulations and labels change continuously — the pipeline must re-validate every month, or the system quietly rots.

This is applied data engineering in service of safety — deterministic, versioned, and auditable end to end.

Success Metrics — Recall Comes First

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A missed allergen reaches a patient. A false alarm costs a moment of review. The metric priorities follow.

Recall

of allergen detection — the safety metric; false negatives are the harm case

Precision

of flags raised — protects clinician trust; too many false alarms and alerts get ignored

Coverage

% of oral medications the pipeline can parse and classify each month

Explainability

every flag names the exact ingredient and source — no black boxes

Alternative rate

% of flagged medications for which Sanjeevini finds a formulary-supported alternative

Freshness

days between openFDA update and system re-validation

Same evaluation discipline as my other projects: a golden dataset, hand-verified labels, and honest reporting of misses.

Named early, managed deliberately — a plan is only credible with its risks attached.

Data dependency	The system inherits openFDA/DailyMed quality and update cadence.	Mitigation: Monthly re-validation; freshness metric; multi-source cross-checks.
Synonym coverage	A missing alias = a missed allergen = patient harm.	Mitigation: Curated dictionary validated against research lists; recall-first evaluation; continuous additions.
Clinical liability	Decision support must never impersonate medical advice.	Mitigation: Human-in-the-loop always: flags inform, clinicians decide; explicit not-medical-advice scope.
Scope perception	Rule-based core may be judged 'not ML enough' academically.	Mitigation: Deliberate choice, stated proudly: for safety, deterministic and auditable is the feature.

Extensibility — One Engine, Many Protections

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The same synonym-aware labelling engine extends far beyond the top-9 allergens.

Celiac disease

Gluten hides in medications too — same detection, different dictionary.

Alpha-gal syndrome

Gelatin capsules and mammalian-derived excipients — a growing, underserved population.

Vegan & religious needs

Gelatin, lactose, and animal-derived ingredients matter to millions beyond allergy.

Pediatric safety

Children's formulations and dosing intersect with the same excipient risks.

Akshaya Patra synergy

Food labels and drug labels are sibling problems — one family of labelling engines, shared synonym infrastructure.

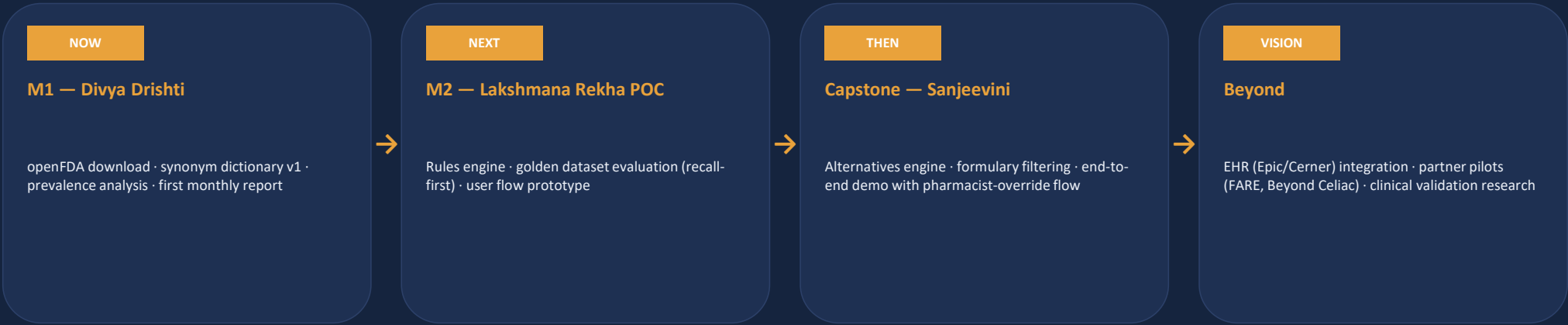
Policy support

Divya Drishti's monthly numbers give the ADINA Act's advocates the evidence base they currently lack.

Roadmap — MADS Milestones to Capstone

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Phased like everything I run: one current focus, gates between phases, parked scope stays parked.



Status: research & problem formulation complete (SIADS 501) · openFDA data acquired · M1 analysis in design

Support I'm seeking: clinical advisors (allergy/immunology), pharmacy database access, and patients willing to test — conversations open with FARE and Beyond Celiac.

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See the problem. Draw the line. Bring the remedy — in time.

No patient should spend four months and \$30,000 discovering what their own records already knew. The data exists. The connection is the product.

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Questions & collaboration welcome.